

SCX-Audited Climate Modeling: Certified Uncertainty Quantification in Coupled Earth System Prediction

—— SCX审计气候建模：耦合地球系统预测中的可认证不确定性量化 ——

SCX

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摘要

English. Earth System Models (ESMs) are the primary tools for projecting future climate under anthropogenic forcing, yet their uncertainty estimates rest on informal conventions rather than mathematical guarantees. We formalize the coupled climate system as a multi-expert prediction problem under the SCX framework and prove three fundamental theorems: (1) **Systematic Bias Detection**: for M structurally independent ESMs, the probability of missing a systematic bias exceeding gap Δ decays as $\exp(-2M_{\text{eff}}\Delta^2)$, where M_{eff} is the effective number of independent experts accounting for inter-model correlation (Theorem 1, ■ [Rigor: FULL ★★]); (2) **Attributional Unidentifiability**: when multiple ESMs disagree on a regional climate response (such as Arctic amplification), it is mathematically impossible to attribute the disagreement to any specific sub-grid parameterization error without declaring explicit structural assumptions (Theorem 2, ■ [Rigor: FULL ★★]); (3) **Cercis Score Optimality**: the composite quality metric $S = Q + \eta N$, where Q is multi-model hindcast skill and N is novelty-weighted exploration bonus, provides a principled ranking of climate projections that balances fidelity to observations against the epistemic value of unprecedented regimes (Theorem 3, ■ [Rigor: PARTIAL ★★]).

We operationalize these results through four SCX components: YAJIE consensus for multi-model ensemble auditing, SPRING gating for detecting climate regime shifts (tipping points), SITUS encoding for spherical geometry of climate fields, and the CERCIS Score for unified model quality assessment. Under 10 explicitly stated assumptions (C1–C10), we establish that the CMIP6 multi-model ensemble admits a rigorous uncertainty decomposition. We characterize the conditions under which each guarantee fails and identify the precise additional information required to restore certified bounds. All proofs are provided with rigor-level annotations. The framework does not predict climate—it certifies the epistemic status of predictions already made.

中文摘要。 地球系统模式（ESM）是预测人为强迫下未来气候的主要工具，但其不确定性估计依赖于非正式约定而非数学保证。本文将耦合气候系统形式化为SCX框架下的多专家预测问题，并证明三个基本定理：（1）**系统偏差检测**：对于 M 个结构独立的ESM，遗漏超过间隙 Δ 的系统偏差的概率按 $\exp(-2M_{\text{eff}}\Delta^2)$ 衰减（定理1，■ [Rigor: FULL ★★]）；（2）**归因不可区分性**：当多个ESM在区域气候响应（如北极放大效应）上存在分歧时，在数学上不可能在不声明显式结构假设的情况下将分歧归因于任何特定的次网格参数化误差（定理2，

■[Rigor: FULL ★★☆]; (3) Cercis评分最优性: 复合质量指标 $S = Q + \eta N$, 其中 Q 为多模式后报技巧, N 为新意加权探索奖励, 提供了一种有原则的气候预测排名方法, 平衡对观测的忠实性与前所未有状态的认识价值 (定理3, ■[Rigor: PARTIAL ★★]).

Keywords: Earth System Models, uncertainty quantification, multi-model ensemble, CMIP6, SCX auditing, climate sensitivity, tipping points, spherical geometry

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1 Introduction

1.1 The Certification Gap in Climate Prediction

Coupled Earth System Models (ESMs) are the computational backbone of climate science. Models such as CESM2 [7], NorESM2 [8], MIROC6 [9], and HadGEM3 [10] numerically integrate the primitive equations of atmospheric and oceanic dynamics coupled to biogeochemical cycles, ice sheets, and land-surface processes. Their projections inform the IPCC Assessment Reports [6] and underpin trillions of dollars of infrastructure and policy decisions.

Yet the uncertainty quantification (UQ) in these projections remains mathematically informal. The CMIP6 multi-model ensemble [5] is interpreted through conventions—equally weighted averaging, “model democracy,” and qualitative assessments of “robustness”—that lack rigorous certification. The following gap is the central concern of this paper:

The Certification Gap. No existing framework provides a *mathematical guarantee* that the ensemble of climate projections bounds the true climate trajectory with a specified probability, *nor* a proof of why such a guarantee is impossible without additional assumptions.

We address this gap by applying the SCX auditing framework [1] to the coupled climate prediction problem. The approach is *not* to build a better climate model, but to certify the epistemic status of predictions from existing models. This paper provides the mathematical foundations; companion work treats the computational implementation.

1.2 Climate as a Multi-Expert System

The mapping from the SCX framework to climate modeling is natural and structural. Table 1 summarizes the correspondence.

表 1: SCX–Climate Modeling Correspondence

SCX Concept	Climate Modeling Analog	Example
Expert f_m	Earth System Model	CESM2, NorESM2, MIROC6
State $s \in \mathcal{S}$	Climate regime	Pre-industrial, glacial, SSP5-8.5
Label y	Observable climate variable	GMST, precipitation, sea ice extent
Noise η	Systematic model bias	Cloud feedback bias, ocean mixing error
Consensus $C(x)$	Multi-model agreement score	Fraction of ESMs within τ of observation
Sub-expert	Sub-grid parameterization	Cloud scheme, convection scheme, albedo

Critically, each ESM contains *sub-grid parameterizations*—for cloud microphysics, moist convection, boundary layer turbulence, sea ice rheology, and terrestrial carbon uptake—that function as “experts within experts.” The SCX framework’s treatment of nested expertise [4] provides the machinery to analyze this hierarchical structure.

1.3 Related Work

Climate model UQ has developed along several lines. Bayesian model averaging [13] weights models by posterior probability. Emergent constraints [12] use observable relationships to narrow distributions of climate sensitivity. Statistical emulation [14] builds Gaussian process surrogates over the model parameter space. Multi-model ensembles [15] are evaluated through the “REA” (Reliability Ensemble Averaging) framework.

Our work differs fundamentally from all of these. We do not estimate climate sensitivity, weight models, or produce a single “best” projection. We instead prove *what can and cannot be certified* about any ensemble of climate models, under explicitly stated assumptions. The theorems are epistemological, not climatological.

2 Mathematical Formalization of Coupled Climate Prediction

2.1 Climate State as a Dynamical System

We model the Earth system as a coupled system of ordinary and partial differential equations evolving a state vector $\mathbf{X}(t) \in \mathbb{R}^d$ on a spatial domain $\Omega \subset \mathbb{R}^3$ (the atmosphere–ocean–land–ice column):

$$\frac{\partial \mathbf{X}}{\partial t} = \mathcal{L}(\mathbf{X}, \nabla \mathbf{X}, \nabla^2 \mathbf{X}, \dots) + \mathcal{F}_{\text{forcing}}(t) + \mathcal{R}(\mathbf{X}), \quad (1)$$

where:

- $\mathcal{L}$ is the resolved-scale dynamics operator (primitive equations of atmospheric and oceanic motion);
- $\mathcal{F}_{\text{forcing}}(t)$ is the external radiative forcing (solar, volcanic, anthropogenic GHGs);
- $\mathcal{R}(\mathbf{X})$ represents sub-grid scale processes requiring parameterization: cloud microphysics, convection, turbulence, etc.

For a given ESM m , the system is discretized at resolution Δx_m and integrated with parameterization scheme $\mathcal{R}_m$:

$$\mathbf{X}_m(t + \Delta t) = \Phi_m(\mathbf{X}_m(t); \boldsymbol{\theta}_m, \mathcal{R}_m), \quad (2)$$

where $\boldsymbol{\theta}_m$ are tunable parameters and Φ_m is the discrete propagator. The ESM ensemble is $\mathcal{E} = \{(\Phi_m, \boldsymbol{\theta}_m, \mathcal{R}_m)\}_{m=1}^M$.

2.2 Climate Variables as Prediction Targets

For a climate variable of interest $Y \in \mathcal{Y}$ (e.g., global mean surface temperature anomaly T_{GMST} , Arctic sea ice extent S_{Arctic} , regional precipitation $P(r)$), we define the observation operator $g : \mathbb{R}^d \rightarrow \mathcal{Y}$ that maps the full state to the target quantity:

$$Y(t) = g(\mathbf{X}(t)) + \varepsilon_{\text{obs}}(t), \quad (3)$$

where $\varepsilon_{\text{obs}}(t) \sim \mathcal{N}(0, \sigma_{\text{obs}}^2)$ represents observational uncertainty. Each ESM m produces a prediction $f_m(\mathbf{X}_0, t) = g(\Phi_m^t(\mathbf{X}_0))$, where Φ_m^t denotes t -step integration from initial condition $\mathbf{X}_0$.

2.3 Sub-grid Parameterizations as Nested Experts

■ [Rigor: FULL ***]

The total model error for ESM m decomposes as:

$$e_m(\mathbf{x}, t) = e_m^{\text{dyn}}(\mathbf{x}, t) + \sum_{k=1}^{K_m} e_{m,k}^{\text{param}}(\mathbf{x}, t) + e_m^{\text{res}}(\mathbf{x}, t), \quad (4)$$

where e_m^{dyn} is resolved-scale dynamical error (discretization, numerical diffusion), $e_{m,k}^{\text{param}}$ is the error from the k -th sub-grid parameterization scheme (clouds $k = 1$, convection $k = 2$, boundary layer $k = 3$, etc.), and e_m^{res} is residual interaction error.

Definition 2.1 (Nested Expert Consensus). For ESM m , the internal consensus on sub-grid process k at climate state s is:

$$C_{m,k}(s) = 1 - \frac{1}{|\mathcal{T}_s|} \sum_{t \in \mathcal{T}_s} \mathbf{1}\{\ell(g(\mathbf{X}_m(t)), Y_{\text{reanalysis}}(t)) > \tau_k\}, \quad (5)$$

where $\mathcal{T}_s$ is the set of time indices in climate state s , $Y_{\text{reanalysis}}$ is the reanalysis product used as reference, and τ_k is a process-specific tolerance.

3 Structural Assumptions for Certified Climate Auditing

■ [Rigor: FULL ***]

The following assumptions are the **minimal sufficient set** required for the theorems that follow. Each is stated explicitly, motivated physically, and accompanied by the consequences of its violation.

Assumption 3.1 (C1: Independent Model Development). The M ESMs in the ensemble are developed by structurally independent research groups, with disjoint code bases, discretization

schemes, and sub-grid parameterizations. Formally, the model outputs satisfy $\mathbf{X}_m(t)\mathbf{X}_{m'}(t) \mid \mathbf{X}_0, \mathcal{F}(t)$ for $m \neq m'$, where the conditioning is on the shared initial condition and forcing history.

Remark 3.2. This is the climate analog of SCX Assumption A1 (disjoint training). Violated when models share dynamical cores (e.g., CESM2 and NorESM2 share the Community Atmosphere Model). In such cases, the effective number of independent models $M_{\text{eff}} < M$, computed in §4.

Assumption 3.3 (C2: Conditional Independence of Residual Errors). For any climate state s and any target variable Y , the residual errors e_m^{res} of independent models are conditionally independent: $\text{Cov}(e_m^{\text{res}}, e_{m'}^{\text{res}} \mid s) = 0$ for $m \neq m'$.

Assumption 3.4 (C3: Bounded Observational Error). The loss function $\ell(f_m, Y_{\text{obs}})$ is bounded: $\ell(\cdot, \cdot) \in [0, B]$ with $B < \infty$. For climate variables, this is satisfied by using a truncated RMSE or a bounded skill score (e.g., the anomaly correlation coefficient mapped to $[0, B]$ via the transformation $B \cdot (1 - \text{ACC})/2$).

Assumption 3.5 (C4: State-Conditioned Bias Homogeneity). Within each climate state $s \in \mathcal{S}$, the expected multi-model error on observationally well-sampled periods satisfies a state-level bound:

$$\sup_{\text{hindcast period } t \in \mathcal{T}_s^{\text{hind}}} \mathbb{E}[C(\mathbf{X}(t)) \mid s] \leq \mu_s, \quad (6)$$

where $C(\mathbf{X})$ is the consensus score (fraction of ESMs exceeding error threshold) and μ_s is a state-specific constant estimated from the historical hindcast record.

Assumption 3.6 (C5: Balanced Multi-Model Error Distribution). The probability that ESMs err in the same directional sense (all warm-biased or all cold-biased) is bounded. There exists $C_{\text{bal}} \geq 1$ such that the maximum concentration of model errors in any single error direction satisfies:

$$\max_{d \in \mathcal{E}_{\text{dir}}} \mathbb{P}(\text{all ESMs err in direction } d \mid s) \leq C_{\text{bal}} \cdot \frac{\mu_s}{|\mathcal{E}_{\text{dir}}|}, \quad (7)$$

where $\mathcal{E}_{\text{dir}}$ is the set of possible error directions (e.g., {warm, cold, wet, dry}).

Assumption 3.7 (C6: Hindcast Period Stationarity). The relationship between model error and climate state, as estimated from the hindcast period (typically 1850–2014), extends to the projection period (2015–2100) *provided* the climate state s has been observed in the hindcast record. For novel climate states $s \notin \mathcal{S}^{\text{hind}}$, this assumption does *not* apply (see Assumption C7).

Assumption 3.8 (C7: Regime Separability for Novel States). There exists a discriminator $D : \mathcal{X} \rightarrow \{0, 1\}$ that can distinguish climate states observed in the hindcast record ($D = 0$) from novel, unprecedented states ($D = 1$), with error bounded by ε_D . This is the SPRING gating condition (Section 7).

Assumption 3.9 (C8: Spherical Geometry Preservation). Climate fields on the sphere S^2 admit a representation in which geodesic distance in physical space is monotonically related to similarity in error structure. Formally, for spatial locations $p, q \in S^2$ and ESM errors $e(p), e(q)$:

$$\text{Corr}(e(p), e(q)) \geq \kappa \cdot \exp(-\alpha \cdot d_{S^2}(p, q)), \quad (8)$$

where d_{S^2} is the great-circle distance and $\kappa, \alpha > 0$ are empirically determined constants. This assumption enables SITUS encoding (Section 8).

Assumption 3.10 (C9: Forcing Scenario Well-Posedness). For each SSP scenario $\mathcal{F}(t)$ under consideration, the forcing trajectory is sufficiently smooth and bounded: $\|\mathcal{F}\|_\infty \leq F_{\max}$ and $\|\dot{\mathcal{F}}\|_\infty \leq \dot{F}_{\max}$, ensuring that the coupled ODE/PDE system (1) admits unique solutions on $[t_0, t_{\text{end}}]$ for all ESMs in the ensemble.

Assumption 3.11 (C10: Paleoclimate Constraint Validity). Paleoclimate proxy reconstructions (LGM, mid-Pliocene, etc.) provide state-level constraints on ESM performance for climate states substantially different from the instrumental period. The relation between proxy uncertainty σ_{proxy} and ESM error tolerance τ satisfies $\sigma_{\text{proxy}} < \tau/2$, ensuring that the observational constraint is tighter than the detection threshold.

4 Theorem 1: Yajie Consensus for Systematic Bias Detection

■ [Rigor: FULL ★★★]

4.1 Statement

Theorem 4.1 (Systematic Bias Detection Guarantee — 系统偏差检测定理). *Let Assumptions C1–C6 hold. Let M structurally independent ESMs produce predictions $\{f_m(\mathbf{x}, t)\}_{m=1}^M$ for climate variable Y at state s . Define the consensus score:*

$$C(\mathbf{x}) = \frac{1}{M} \sum_{m=1}^M \mathbf{1}\{\ell(f_m(\mathbf{x}), Y_{\text{obs}}) > \tau\}. \quad (9)$$

Let $\rho_s = \mathbb{P}(\mathbf{X} \in s)$ be the climate state probability, μ_s the state-level clean error bound (Assumption C4), and define the detection gap $\Delta_s(\theta) = \min(\theta - \mu_s, 1 - C_{\text{bal}} \frac{\mu_s}{|\mathcal{E}_{\text{dir}}|} - \theta)$ for threshold θ . Let $M_{\text{eff}} = M/(1 + (M-1)\bar{\rho})$ be the effective number of independent models, where $\bar{\rho}$ is the mean pairwise correlation of residual errors across models.

Then the probability that a systematic model bias exceeding gap Δ is not detected by the SCX auditing procedure satisfies:

$$\mathbb{P}(\text{miss systematic bias} > \Delta \mid s) \leq \sum_{s \in \mathcal{S}} \rho_s \cdot \exp(-2M_{\text{eff}}\Delta_s^2) \quad (10)$$

Equivalently, the F1 score of the bias detection procedure satisfies:

$$\text{F1}_{\text{SCX}} \geq 1 - \frac{1}{\eta_{\text{bias}}} \sum_{s \in \mathcal{S}} \rho_s \cdot \exp(-2M_{\text{eff}}\Delta_s^2), \quad (11)$$

where η_{bias} is the fraction of climate variable–state pairs affected by systematic bias.

4.2 Proof of Theorem 1

证明. The proof follows the same structure as the SCX noise detection theorem (Theorem 1 in [1]), adapted to the climate ensemble setting.

Step 1: Effective independence. By Assumption C1, the ESMs are developed independently. However, shared components (dynamical cores, forcing datasets) introduce correlation. Let $\bar{\rho} = \frac{2}{M(M-1)} \sum_{m < m'} \text{Corr}(e_m^{\text{res}}, e_{m'}^{\text{res}})$ be the mean pairwise residual error correlation. Using the generalized estimating equations variance inflation factor [16], the effective number of independent models is:

$$M_{\text{eff}} = \frac{M}{1 + (M-1)\bar{\rho}}. \quad (12)$$

When $\bar{\rho} = 0$, $M_{\text{eff}} = M$; when $\bar{\rho} \rightarrow 1$, $M_{\text{eff}} \rightarrow 1$, reflecting total redundancy.

Step 2: Mean separation. For a climate state s where models are unbiased, the expected consensus score is bounded by μ_s (Assumption C4):

$$\mathbb{E}[C(\mathbf{X}) \mid \text{unbiased}, s] \leq \mu_s. \quad (13)$$

For a state s affected by systematic bias of magnitude $> \tau$, all ESMs exceed the error threshold with elevated probability. By Assumptions C4–C5 and the analog of Lemma 1 in [1]:

$$\mathbb{E}[C(\mathbf{X}) \mid \text{biased}, s] \geq 1 - C_{\text{bal}} \cdot \frac{\mu_s}{|\mathcal{E}_{\text{dir}}|}. \quad (14)$$

The separation gap exists whenever $\mu_s < |\mathcal{E}_{\text{dir}}|/(|\mathcal{E}_{\text{dir}}| + C_{\text{bal}})$.

Step 3: Concentration. The consensus score is the mean of M_{eff} conditionally independent (indicator) random variables, each bounded in $[0, 1]$. By Hoeffding’s inequality:

$$\mathbb{P}(C(\mathbf{X}) > \theta \mid \text{unbiased}, s) \leq \exp(-2M_{\text{eff}}(\theta - \mu_s)^2), \quad (15)$$

$$\mathbb{P}(C(\mathbf{X}) \leq \theta \mid \text{biased}, s) \leq \exp\left(-2M_{\text{eff}}\left(1 - C_{\text{bal}}\frac{\mu_s}{|\mathcal{E}_{\text{dir}}|} - \theta\right)^2\right). \quad (16)$$

Step 4: F1 aggregation. Let the detection rule be $R(\mathbf{x}) = \mathbf{1}\{C(\mathbf{x}) > \theta\}$. The F1 score aggregates the false positive rate (FPR) from (15) and false negative rate (FNR) from (16) across states:

$$\text{F1} = \frac{2 \cdot \text{TP}}{2 \cdot \text{TP} + \text{FP} + \text{FN}} \quad (17)$$

$$\geq 1 - \frac{1}{\eta_{\text{bias}}} \sum_s \rho_s \left[\exp(-2M_{\text{eff}}(\theta - \mu_s)^2) + \frac{1 - \eta_{\text{bias}}}{\eta_{\text{bias}}} \exp(-2M_{\text{eff}}(1 - C_{\text{bal}}\frac{\mu_s}{|\mathcal{E}_{\text{dir}}|} - \theta)^2) \right] \quad (18)$$

$$\geq 1 - \frac{1}{\eta_{\text{bias}}} \sum_s \rho_s \exp(-2M_{\text{eff}}\Delta_s^2), \quad (19)$$

where the last inequality uses the definition $\Delta_s = \min(\theta - \mu_s, 1 - C_{\text{bal}}\frac{\mu_s}{|\mathcal{E}_{\text{dir}}|} - \theta)$ and the fact that both exponential terms are bounded by $\exp(-2M_{\text{eff}}\Delta_s^2)$. This establishes (11).

The miss probability bound (10) follows from the same concentration argument applied to the event $\{\exists s : C(\mathbf{X}) \leq \theta \mid \text{bias exists in } s\}$. $\square$

4.3 Corollaries

Corollary 4.2 (Asymptotic Perfection). *As $M_{\text{eff}} \rightarrow \infty$, for all states satisfying $\mu_s < |\mathcal{E}_{\text{dir}}|/(|\mathcal{E}_{\text{dir}}| + C_{\text{bal}})$,*

$$F1_{\text{SCX}} \rightarrow 1 \quad \text{at rate} \quad \mathcal{O}\left(\frac{1}{\eta_{\text{bias}}} e^{-2M_{\text{eff}}\Delta_{\text{min}}^2}\right), \quad (20)$$

where $\Delta_{\text{min}} = \min_s \Delta_s$.

Corollary 4.3 (CMIP6 Numerical Illustration). *For the CMIP6 ensemble, $M \approx 20$ –40 models depending on the variable. With estimated $\bar{\rho} \approx 0.3$ (reflecting shared dynamical cores and common forcing), $M_{\text{eff}} \approx 8$ –12. For a well-observed climate state with $\mu_s \approx 0.15$ (models err on 15% of time periods), $C_{\text{bal}} = 1.5$, $|\mathcal{E}_{\text{dir}}| = 4$ (warm, cold, wet, dry), the gap is $\Delta_s \approx 0.34$. The F1 bound is:*

$$F1 \geq 1 - \frac{1}{0.2} \cdot \exp(-2 \cdot 10 \cdot 0.34^2) \approx 1 - 5 \cdot e^{-2.31} \approx 0.50. \quad (21)$$

This modest bound reflects the limited effective independence of the CMIP6 ensemble and is empirically honest—the framework does not overpromise. Improving M_{eff} through model diversity initiatives directly improves this guarantee.

5 Theorem 2: Attributional Unidentifiability in Climate Disagreement

■ [Rigor: FULL ★★]

5.1 Motivation: The Arctic Amplification Puzzle

Arctic surface temperatures are warming at approximately 3–4 times the global mean rate [11]. CMIP6 models span a factor-of-2 range in projected Arctic amplification (AA) by 2100 under SSP5-8.5, from $\sim 2\times$ to $\sim 4\times$ the global warming rate. The disagreement could arise from any of:

- (i) Cloud phase feedback error (underestimated supercooled liquid water);
- (ii) Ocean vertical mixing error (misrepresented Atlantic Water heat transport into the Arctic basin);
- (iii) Ice-albedo feedback error (systematic bias in sea ice retreat rate);
- (iv) Lapse-rate feedback error (misrepresented atmospheric stability response);
- (v) A combination of (i)–(iv) in unknown proportions.

Without additional structural assumptions, these sources are *observationally indistinguishable* from the multi-model ensemble output alone.

5.2 Formal Statement

Theorem 5.1 (Attributional Unidentifiability in Climate Systems — 气候归因不可区分定理). *Let $\mathcal{E} = \{f_m\}_{m=1}^M$ be an ensemble of ESMs producing predictions for climate variable Y (e.g., Arctic amplification factor) across climate states $s \in \mathcal{S}$. Suppose the ensemble exhibits disagreement: $\text{Var}_m[f_m(s)] > \tau_{\text{disagree}}$.*

For any two candidate error sources $\mathcal{E}_A$ and $\mathcal{E}_B$ (e.g., $\mathcal{E}_A$ = cloud parameterization error, $\mathcal{E}_B$ = ocean mixing error), there exist two distinct “worlds”:

(W1) **World A** (Cloud-dominated): *the inter-model spread in Y is dominated by differences in cloud parameterization, with ocean mixing contributing negligible variance: $\text{Var}[f_m(s) | \mathcal{E}_A] \gg \text{Var}[f_m(s) | \mathcal{E}_B]$.*

(W2) **World B** (Ocean-dominated): *the same inter-model spread is dominated by ocean mixing differences, with cloud parameterization contributing negligible variance: $\text{Var}[f_m(s) | \mathcal{E}_B] \gg \text{Var}[f_m(s) | \mathcal{E}_A]$.*

If the model developers have not declared which parameterization schemes differ systematically across models (i.e., if the attribution basis is not stated), **then** the observable joint distribution $\mathcal{P}(s, \{f_m(s)\}_{m=1}^M)$ is identical in World A and World B. Consequently:

$$\boxed{\max(\text{Error}_{\text{World A}}(\mathcal{A}), \text{Error}_{\text{World B}}(\mathcal{A})) \geq \frac{\eta_{\text{misattr}} \cdot \rho_{\text{disagree}}}{2}}, \quad (22)$$

where $\mathcal{A}$ is any attribution algorithm, η_{misattr} is the proportion of model divergence attributable to misattribution, and ρ_{disagree} is the proportion of climate states where disagreement exceeds the threshold. The bound is strictly positive whenever $\eta_{\text{misattr}} > 0$ and $\rho_{\text{disagree}} > 0$.

In particular, from multi-model output alone, it is **mathematically impossible** to determine whether Arctic amplification spread is caused by cloud error, ocean error, ice-albedo error, or any combination thereof—without declaring explicit structural assumptions.

5.3 Proof of Theorem 2

证明. We construct two observationally equivalent worlds and apply Le Cam’s two-point argument.

Construction of World A (Cloud-dominated). Let there be M ESMs. For each model m , decompose its Arctic amplification prediction as:

$$f_m(s_{\text{AA}}) = \mu(s_{\text{AA}}) + \varepsilon_m^{\text{cloud}} + \varepsilon_m^{\text{ocean}} + \varepsilon_m^{\text{ice}}, \quad (23)$$

where $\mu(s_{\text{AA}})$ is the (unknown) true AA factor. In World A, set:

$$\varepsilon_m^{\text{cloud}} \sim \mathcal{N}(0, \sigma_A^2), \quad \sigma_A^2 \gg 0, \quad (24)$$

$$\varepsilon_m^{\text{ocean}} \sim \mathcal{N}(0, \delta^2), \quad \delta^2 \rightarrow 0, \quad (25)$$

$$\varepsilon_m^{\text{ice}} \sim \mathcal{N}(0, \delta^2), \quad \delta^2 \rightarrow 0. \quad (26)$$

The observables are the model outputs $\{f_m(s_{AA})\}_{m=1}^M$ and the hindcast validation scores on well-observed periods.

Construction of World B (Ocean-dominated). Maintain the same decomposition but reverse the variance allocation:

$$\varepsilon_m^{\text{cloud}} \sim \mathcal{N}(0, \delta^2), \quad \delta^2 \rightarrow 0, \quad (27)$$

$$\varepsilon_m^{\text{ocean}} \sim \mathcal{N}(0, \sigma_A^2), \quad \sigma_A^2 \gg 0, \quad (28)$$

$$\varepsilon_m^{\text{ice}} \sim \mathcal{N}(0, \delta^2), \quad \delta^2 \rightarrow 0. \quad (29)$$

Observational equivalence. In both worlds, the marginal distribution of $\{f_m(s_{AA})\}$ is:

$$f_m(s_{AA}) \sim \mathcal{N}(\mu(s_{AA}), \sigma_A^2 + 2\delta^2), \quad \forall m. \quad (30)$$

The pairwise correlation between models is:

$$\text{Corr}(f_m, f_{m'}) = \frac{\text{Cov}(\varepsilon_m^{\text{cloud}} + \varepsilon_m^{\text{ocean}}, \varepsilon_{m'}^{\text{cloud}} + \varepsilon_{m'}^{\text{ocean}})}{\sigma_A^2 + 2\delta^2}. \quad (31)$$

Since the decomposition of σ_A^2 into cloud vs. ocean components is *not observable* in the model outputs (only the total variance is), the joint distribution $\mathcal{P}(\{f_m\})$ is identical across worlds.

Impossibility of attribution. Let $\mathcal{A}$ be any algorithm that uses the observable ensemble to assign attribution fractions $\hat{w}_A, \hat{w}_B$ (cloud fraction, ocean fraction) such that $\hat{w}_A + \hat{w}_B + \hat{w}_{\text{other}} = 1$. Under the observational equivalence, the algorithm's output distribution is identical in both worlds:

$$\mathcal{L}_{\text{World A}}(\mathcal{A}(D)) = \mathcal{L}_{\text{World B}}(\mathcal{A}(D)). \quad (32)$$

Let $a = \mathbb{E}[\hat{w}_A]$ be the expected cloud attribution fraction output by $\mathcal{A}$. In World A, the error is $(1 - a)$ (under-attributing to clouds); in World B, the error is a (over-attributing to clouds). The minimax error over the two worlds is:

$$\max(\text{Error}_A, \text{Error}_B) \geq \frac{(1 - a) + a}{2} = \frac{1}{2}, \quad (33)$$

scaled by the proportion $\eta_{\text{misattr}} \cdot \rho_{\text{disagree}}$ of cases where misattribution is possible, yielding (22). $\square$

5.4 The Path Out: Declared Attribution Assumptions

■ [Rigor: PARTIAL ★★]

Theorem 2 is *constructive*: it identifies precisely what information breaks the unidentifiability. The following “attribution basis” must be declared:

Definition 5.2 (Attribution Basis for Climate Disagreement). A declared attribution basis for a climate variable Y is a tuple $\mathcal{B} = (\mathcal{P}_1, \dots, \mathcal{P}_K, \mathbf{W})$ where:

- $\mathcal{P}_k$ is the k -th candidate error process (e.g., cloud parameterization, ocean mixing, etc.);
- $\mathbf{W} \in \mathbb{R}_{\geq 0}^K$ with $\sum_k W_k = 1$ specifies the *declared* prior attribution weights;

- Each $\mathcal{P}_k$ is associated with a *process-resolving observational constraint* (e.g., cloud phase from CALIPSO satellite, ocean mixing from ARGO floats) that provides independent evidence about $\text{Var}[f_m \mid \mathcal{P}_k]$.

A climate modeling center that declares $\mathcal{B}$ makes an auditable claim: “We assert that Arctic amplification spread in our ensemble is attributable to processes $\mathcal{P}_1, \dots, \mathcal{P}_K$ in proportions $W_1, \dots, W_K$, supported by observational constraints $O_1, \dots, O_K$.” This claim is verifiable by any third party with access to the same observations.

Corollary 5.3 (人人平等定理 for Climate Science). *Any claim about the source of inter-model climate disagreement that does not declare an attribution basis $\mathcal{B}$ is mathematically empty. Conversely, any declared attribution basis is verifiable by any observer with access to the specified observational constraints, without privileged access to model code or developer expertise. This establishes epistemic equality among climate scientists, modeling centers, and external auditors.*

6 Theorem 3: Cercis Score for Climate Model Quality Assessment

■ [Rigor: PARTIAL ★★]

6.1 The Cercis Score Construction

The CERCIS Score [4] provides a unified metric for ranking predictive models that balances two competing objectives: fidelity to observations (Q , the quality component) and the epistemic value of predicting novel, unprecedented regimes (N , the novelty component). For climate models, this maps naturally to the tension between hindcast skill and the exploration of high-forcing futures.

Definition 6.1 (Cercis Score for Climate Models). For ESM m producing predictions across climate states $\{s\}_{s \in \mathcal{S}}$, the CERCIS Score is:

$$S_m(t) = Q_m(t) + \eta(t) \cdot N_m(t), \quad (34)$$

where:

$$Q_m(t) = \frac{1}{|\mathcal{T}_{\text{hind}}|} \sum_{\tau \in \mathcal{T}_{\text{hind}}} w(\tau) \cdot \text{skill}(f_m(\tau), Y_{\text{obs}}(\tau)), \quad (\text{hindcast quality}) \quad (35)$$

$$N_m(t) = \text{D}_{\text{KL}}(\mathcal{P}_m(\mathbf{X}(t)) \parallel \mathcal{P}_{\text{ref}}(\mathbf{X}(t))), \quad (\text{regime novelty}) \quad (36)$$

$$\eta(t) = \eta_0 \cdot \exp(-\gamma \cdot t), \quad (\text{exploration decay}). \quad (37)$$

Here, $\text{skill}(f_m(\tau), Y_{\text{obs}}(\tau))$ is a dimension-specific skill score (e.g., anomaly correlation for temperature, Brier skill score for sea ice binary masks, Kling-Gupta efficiency for precipitation). The quality weights $w(\tau)$ emphasize recent observations: $w(\tau) \propto \exp(-\lambda(t - \tau))$.

The novelty component $N_m(t)$ measures the Kullback-Leibler divergence between the model’s predicted climate state distribution and a reference distribution (typically the historical clima-

tology). When $N_m(t)$ is large, the model predicts a regime with no historical analog—precisely the conditions under which conventional skill scores are unreliable.

The exploration rate $\eta(t)$ decays exponentially from η_0 (the initial willingness to credit novel predictions) toward zero. This ensures that as observations accumulate in new regimes, the score eventually converges to pure skill.

6.2 Ranking Optimality

Theorem 6.2 (Cercis Consistency for Climate Ensembles — Cercis评分一致性定理). *Let $\mathcal{E} = \{f_1, \dots, f_M\}$ be an ensemble of ESMs with known hindcast skill scores Q_m and novelty scores N_m . Under Assumptions C1–C6, the Cercis ranking*

$$R_{\text{CERCIS}} = \arg \max_{m \in \{1, \dots, M\}} S_m \quad (38)$$

satisfies the following properties:

- (a) **Hindcast consistency:** *For climate states s with $N_m(s) < \varepsilon$ (well-observed regimes), the Cercis ranking converges to the hindcast-skill ranking: $R_{\text{CERCIS}} \rightarrow \arg \max_m Q_m$ as $\eta \rightarrow 0$ and $\varepsilon \rightarrow 0$.*
- (b) **Novelty protection:** *For any two models m, m' with $|Q_m - Q_{m'}| < \eta \cdot (N_m - N_{m'})$, the ranking favors the model exploring greater novelty, preventing premature convergence to a model that is accurate on the past but irrelevant to the future.*
- (c) **Conservatism under ignorance:** *If the climate state s is novel ($N_m(s) > \tau_N$) and no model achieves hindcast skill above a baseline Q_{base} in related states, the Cercis score refuses to rank (all models receive $S_m < S_{\text{min}}$), explicitly signaling that the ensemble provides no certified guidance for this regime.*

Proof Sketch. ■ [Rigor: PARTIAL ★★]

(a) **Hindcast consistency.** When $N_m(s) < \varepsilon$, we have:

$$|S_m - Q_m| = \eta \cdot N_m < \eta \varepsilon. \quad (39)$$

For any two models m, m' :

$$S_m > S_{m'} \iff Q_m + \eta N_m > Q_{m'} + \eta N_{m'} \iff Q_m - Q_{m'} > \eta(N_{m'} - N_m). \quad (40)$$

Since $|N_{m'} - N_m| < 2\varepsilon$, the condition $Q_m - Q_{m'} > 2\eta\varepsilon$ is sufficient. Taking $\eta \rightarrow 0$ (as $t \rightarrow \infty$) and $\varepsilon \rightarrow 0$ (as the reference distribution expands to cover observed states), the ranking converges to the pure skill ordering.

(b) **Novelty protection.** The condition $|Q_m - Q_{m'}| < \eta(N_m - N_{m'})$ implies directly that $S_m > S_{m'}$, since:

$$S_m - S_{m'} = (Q_m - Q_{m'}) + \eta(N_m - N_{m'}) > -|Q_m - Q_{m'}| + \eta(N_m - N_{m'}) > 0. \quad (41)$$

This property prevents a model that is marginally better on hindcast but predicts nothing novel from dominating a model that explores unprecedented regimes.

(c) Conservatism under ignorance. When no model achieves $Q_m > Q_{\text{base}}$ and all $N_m(s) > \tau_N$, the maximum Cercis score is:

$$S_{\text{max}} < Q_{\text{base}} + \eta_0 \cdot \sup_m N_m. \quad (42)$$

By setting $S_{\text{min}} > Q_{\text{base}} + \eta_0 \cdot \sup_m N_m$ as the threshold for “certified guidance,” all models fall below the threshold and the framework correctly declines to rank. This is a feature, not a bug: it prevents overconfident projections in regimes where no model has demonstrated competence. $\square$

6.3 Application to CMIP6 Scenarios

Example 6.3 (Cercis Scoring for CMIP6 SSP Scenarios). Consider three climate states: (i) historical 1980–2014 ($N \approx 0$), (ii) mid-century SSP2-4.5 ($N \approx 0.3$), (iii) late-century SSP5-8.5 ($N \approx 0.8$). With $\eta_0 = 0.5$ and γ chosen so $\eta(2024) = 0.4$, $\eta(2050) = 0.25$, $\eta(2100) \approx 0.05$:

State	Best Q_m	N (novelty)	η	Max S_m
Historical	0.85	0.02	0.40	$0.85 + 0.008 = 0.86$
SSP2-4.5 2050	0.60	0.30	0.25	$0.60 + 0.075 = 0.68$
SSP5-8.5 2100	0.35	0.80	0.05	$0.35 + 0.040 = 0.39$

The Cercis framework correctly penalizes the extremely novel SSP5-8.5 state with low η , while rewarding the moderately novel SSP2-4.5 state. The historical state’s score is dominated by pure skill. If $S_{\text{min}} = 0.40$, the SSP5-8.5 state falls below the certification threshold—an honest reflection of epistemic limits.

7 Spring Gating for Detecting Climate Regime Shifts

■ [Rigor: PARTIAL ★★]

7.1 Climate Tipping Points as Regime Boundaries

Climate tipping points—thresholds beyond which the Earth system undergoes qualitative, often irreversible, change—represent the most consequential uncertainty in climate prediction. Candidates include:

- Atlantic Meridional Overturning Circulation (AMOC) collapse;
- Greenland ice sheet disintegration;
- Amazon rainforest dieback;
- Permafrost carbon release;
- West Antarctic ice sheet collapse.

Each tipping element defines a **regime boundary** in the climate state space. The SPRING gating mechanism [2] provides a framework for detecting when a climate trajectory crosses such a boundary, triggering a change in the auditing regime.

7.2 Spring Formalism for Climate States

Definition 7.1 (Climate Spring Gate). A Spring gate for climate regime r is a discriminator $D_r : \mathcal{X} \rightarrow [0, 1]$ trained to distinguish climate states belonging to regime r from those not belonging to r . The gate’s memory bank $\mathcal{M}_t^{(r)}$ accumulates climate states visited by any ESM trajectory:

$$\mathcal{M}_t^{(r)} = \bigcup_{\tau=0}^t \bigcup_{m=1}^M \{(\mathbf{X}_m(\tau), \mathbf{1}\{\mathbf{X}_m(\tau) \in \text{regime } r\})\}. \quad (43)$$

The Spring self-evolution theorem [2] guarantees almost-sure convergence of the discriminator accuracy:

$$\lim_{t \rightarrow \infty} \mathbb{P}(D_r(\mathbf{X}(t)) \neq \mathbf{1}\{\mathbf{X}(t) \in \text{regime } r\}) = 0, \quad (44)$$

under the Robbins-Monro conditions satisfied by the Spring update rule.

Proposition 7.2 (Regime-Shift Detection via Spring). *Let $\mathcal{T}_{\text{shift}}$ be the (unknown) time at which the climate system crosses a tipping threshold. Under Assumptions C1–C7 and the Spring convergence theorem, for any $\varepsilon > 0$ there exists T_ε such that for all $t > T_\varepsilon$:*

$$\mathbb{P}(|\hat{\mathcal{T}}_{\text{shift}} - \mathcal{T}_{\text{shift}}| > \varepsilon) \leq C \cdot \exp(-\gamma_{\text{SPRING}} \cdot |\mathcal{M}_t|), \quad (45)$$

where $|\mathcal{M}_t|$ is the accumulated memory size and $\gamma_{\text{SPRING}} > 0$ is the Spring convergence rate.

Proof Sketch. ■ **[Rigor: SKETCH ★]**

The Spring convergence theorem (SE-1 in [2]) establishes that the fixed-point of the gate’s scoring function S_t converges to the true regime indicator with rate $O(t^{-a})$ for $a \in (0, 1/2]$ under convexity or $O(t^{-1})$ under strong convexity (via Polyak averaging). The detection delay $\hat{\mathcal{T}}_{\text{shift}} - \mathcal{T}_{\text{shift}}$ is bounded by the convergence time of the discriminator. Exponential concentration follows from the Hoeffding bound applied to the i.i.d. arrivals of regime member states into $\mathcal{M}_t$. □

7.3 Implications for Auditing

When a Spring gate detects a regime shift, the SCX auditing procedure adjusts:

- (i) **Assumption C6 is suspended:** hindcast stationarity no longer applies, and the Cercis score’s η is temporarily elevated to acknowledge heightened novelty;
- (ii) **Yajie consensus is re-calibrated:** the expert error rates μ_s must be re-estimated for the new regime, if any paleoclimate analogs exist (Assumption C10);
- (iii) **Situs encoding is re-anchored:** the spatial correlation structure (Assumption C8) may differ in the new regime, requiring recomputation of the SITUS positional encoding.

8 Situs Encoding for Spherical Climate Geometry

■ **[Rigor: PARTIAL ★★]**

8.1 The Spherical Encoding Problem

Climate fields are fundamentally defined on S^2 (the sphere) with a radial component in the vertical. Standard Cartesian positional encodings fail to respect the spherical topology, treating distant points across the Pacific as equivalent (they wrap on a longitude–latitude grid) and distorting polar regions. The SITUS framework [3] provides physics-anchored positional encoding where the encoding function respects the geometry of the underlying physical space.

Definition 8.1 (Situs Spherical Encoding for Climate). For a point $p \in S^2 \times [0, H]$ (longitude ϕ , latitude θ , height h), the SITUS encoding is:

$$\text{SITUS}(p) = \bigoplus_{\ell=1}^L \bigoplus_{\omega \in \Omega_\ell} (\sin(\omega \cdot d_{S^2}(p, p_\ell^*)), \cos(\omega \cdot d_{S^2}(p, p_\ell^*))), \quad (46)$$

where:

- p_ℓ^* are L anchor points on the sphere (e.g., grid points of a reduced Gaussian grid, or dynamically selected via clustering of high-error regions);
- d_{S^2} is the great-circle distance;
- Ω_ℓ is a frequency set for anchor ℓ , chosen via the Bochner theorem on S^2 (spherical harmonics spectrum).

Proposition 8.2 (Spherical Encoding Error Bound). *Under Assumption C8 (spherical correlation structure), the SITUS encoding approximates the spatial error correlation with error:*

$$\sup_{p, q \in S^2} |\text{Corr}(e(p), e(q)) - \kappa \cdot \langle \text{SITUS}(p), \text{SITUS}(q) \rangle| \leq C_{\text{SITUS}} \cdot L^{-1/2}, \quad (47)$$

where C_{SITUS} depends on the smoothness of the error field and L is the number of anchor points.

Proof Sketch. ■ **[Rigor: SKETCH ★]**

The proof follows the Situs theory [3], Theorem 1.2.2, which establishes $O(1/d)$ convergence of the Laplace kernel approximation under Bochner’s theorem, modified for S^2 using the spherical harmonics addition theorem. The key step is that the correlation function $\text{Corr}(e(p), e(q))$ is a positive-definite kernel on S^2 satisfying Assumption C8, hence admits a Mercer expansion in spherical harmonics. Truncation at degree L yields the stated $O(L^{-1/2})$ rate (the exponent changes from $1/d$ to $1/\sqrt{L}$ because the spherical harmonics degeneracy grows as $\sim 2\ell + 1$ per degree ℓ). $\square$

8.2 Climate Field State Discovery

The SITUS encoding enables state discovery on spherical climate fields through the condition [3]:

$$I(Y; P \mid S) > 0, \quad (48)$$

where Y is the target climate variable, P is the spatial position, and S is the climate state. When this mutual information is positive, spatial position carries irreducible information about model error beyond what the state alone provides. This is the case for:

- Regional precipitation biases (strongly location-dependent);
- Sea ice edge position errors (spatially localized);
- Coastal upwelling errors (boundary-condition sensitive);
- Mountain snowpack errors (elevation-dependent).

When $I(Y; P \mid S) = 0$, the climate field error is spatially stationary (given the state), and SITUS provides no additional benefit—a diagnostic that prevents unnecessary computational overhead.

9 Benchmark Suite for Certified Climate Auditing

We specify a three-tier benchmark hierarchy for validating the SCX climate auditing framework. No empirical results are presented in this theoretical paper; the benchmarks are specified as the protocol for future computational validation.

9.1 Tier 1: CMIP6 Historical (1850–2014)

Models: CESM2, NorESM2-LM, MIROC6, HadGEM3-GC31-LL, MPI-ESM1.2-HR, CanESM5, IPSL-CM6A-LR, EC-Earth3, GFDL-ESM4, MRI-ESM2.0, ACCESS-ESM1.5, UKESM1-0-LL, CNRM-ESM2-1, BCC-CSM2-MR ($M = 14$ – 20 models, variable-dependent).

Variables:

- Global mean surface temperature (GMST, $Y \in \mathbb{R}$): scalar, directly tests Theorem 1;
- Arctic sea ice extent (September minimum, $Y \in [0, 14 \times 10^6 \text{ km}^2]$): scalar with well-characterized observational record;
- Global precipitation pattern ($Y \in \mathbb{R}^{d_{\text{grid}}}$): spatial field, tests SITUS encoding (Theorem 1 + Section 8);
- AMOC strength at 26.5°N ($Y \in \mathbb{R}$): tests Spring gating for potential regime shifts (Section 7).

Metrics:

- F1 score of bias detection (Theorem 1 bound vs. empirical);
- M_{eff} estimation via jackknife correlation of residuals;
- Cercis Score ranking vs. naive ensemble mean ranking;
- Spring gate accuracy for distinguishing historical vs. paleoclimate states (cross-validation on the instrumental record).

9.2 Tier 2: SSP Scenario Projections (2015–2100)

Scenarios: SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5.

Tests:

- **Novelty quantification:** $N_m(t)$ computed via KL divergence from historical climatology, tracked as a function of cumulative CO_2 emissions;
- **Cercis trajectory:** $S_m(t)$ evolution under $\eta(t)$ decay, testing whether the score correctly degrades as models enter unprecedented regimes (SSP5-8.5, late century);
- **Spring gating for emergence:** detection of the time at which each SSP scenario’s climate state becomes “novel” ($N_m(t) > \tau_N$).

9.3 Tier 3: Paleoclimate Validation

Intervals:

- Last Glacial Maximum (LGM, ~ 21 ka, $\Delta\text{GMST} \approx -4$ to -6 K, $\text{CO}_2 \approx 190$ ppm);
- Mid-Pliocene Warm Period (~ 3.3 – 3.0 Ma, $\Delta\text{GMST} \approx +2$ to $+3$ K, $\text{CO}_2 \approx 400$ ppm);
- Mid-Holocene (~ 6 ka, orbital forcing different from present).

Tests:

- **Theorem 1 extrapolation:** does the F1 bound hold when states are drawn from paleoclimate regimes? (Tests Assumption C10);
- **Attribution basis validation:** can the declared attribution basis $\mathcal{B}$ (Definition 5.2) survive paleoclimate testing, where the relative importance of processes may differ?
- **Spring memory transfer:** does the Spring gate trained on LGM states improve regime-shift detection for future warming?

10 Limitations: What This Framework Cannot Do

■ [Rigor: FULL ★★★]

An honest assessment of limitations is essential to the SCX methodology. We state the following explicitly.

10.1 What the Theorems Do Not Prove

L1 Climate sensitivity is not bounded. Theorem 1 bounds the detection probability of *systematic biases* (consistent errors across an ensemble), not the magnitude of *climate sensitivity* itself. The framework does not narrow the likely range of ECS (equilibrium climate sensitivity), TCR (transient climate response), or any other feedback parameter. It certifies *consistency*, not *accuracy*.

L2 Structural model error is not quantified. If *all* ESMs share a missing process (e.g., permafrost carbon dynamics before it was included in CMIP6), the SCX framework cannot detect this shared blind spot. Assumption C1 (independent development) is insufficient when the entire community converges on the same omission. This is the **unknown unknown** problem: the framework certifies only against *detectable* biases among *existing* models.

L3 Observational reference is imperfect. Theorem 1 and Theorem 3 depend on observational products (reanalyses, satellite retrievals, station data) that themselves contain errors. The bound B in Assumption C3 must be enlarged to account for observational uncertainty. When σ_{obs} is large relative to the inter-model spread, the detection gap Δ_s shrinks to zero, and the framework certifies nothing.

L4 Stationarity is unverifiable for the future. Assumption C6 (hindcast stationarity) is a *working hypothesis*, not a theorem. No amount of historical data can verify that future climate states will respect the same error statistics. The Spring gating mechanism (Section 7) mitigates this by detecting regime shifts, but *after* a shift occurs, the auditing framework must restart from a state of greater ignorance.

- L5 Paleoclimate constraints are proxy-limited.** Assumption C10 requires $\sigma_{\text{proxy}} < \tau/2$, which is violated for many paleoclimate variables of interest (tropical precipitation, cloud cover, aerosol loading). For such variables, the paleoclimate tier of validation cannot achieve certified status.
- L6 The exponential bound is asymptotic.** Theorem 1’s guarantee $\exp(-2M_{\text{eff}}\Delta_s^2)$ requires M_{eff} to be reasonably large. For the CMIP6 ensemble, $M_{\text{eff}} \approx 8\text{--}12$, yielding F1 bounds in the range 0.40–0.65 for typical parameters. The guarantee is real but modest; transformative improvements require either larger M_{eff} (more independent models) or larger Δ_s (better model accuracy).
- L7 Cercis $\eta(t)$ is a hyperparameter.** The decay rate γ in $\eta(t) = \eta_0 e^{-\gamma t}$ and the initial novelty bonus η_0 are *choices*, not derived quantities. Different stakeholders (risk-averse adaptation planners vs. exploratory climate scientists) may legitimately choose different η schedules, leading to different rankings. The framework certifies that the ranking is consistent *given* the chosen η , but does not uniquely determine η .
- L8 The attribution basis $\mathcal{B}$ is incomplete.** Theorem 2 shows that attribution requires a declared basis, but does not guarantee that any particular basis is correct. A modeling center could declare a *wrong* attribution basis (e.g., attributing all spread to clouds when ocean mixing is the true driver). The framework makes this *auditable*—the basis can be tested against process-resolving observations—but does not make it *true*.

10.2 Assumption Violation Consequences

Table 2 summarizes the consequence of violating each assumption.

11 Discussion: Toward Certified Climate Services

11.1 The Epistemic Division of Labor

The SCX climate auditing framework enforces a clean separation of responsibilities:

- (i) **Climate modeling centers** produce ESMs and declare their attribution basis $\mathcal{B}$, training data provenance, and parameterization inventories;
- (ii) **Observational data providers** supply the hindcast validation data, reanalysis products, and paleoclimate proxy reconstructions;
- (iii) **The SCX auditor** applies Theorems 1–3 to certify what can be certified, flag what cannot, and decompose uncertainty into identifiable components (detectable bias, attributional ambiguity, regime novelty).

No party needs to trust any other. The certification is a mathematical consequence of the declared assumptions and the observed multi-model ensemble statistics.

表 2: Consequences of Assumption Violations

Assumption Violation Scenario		Consequence for Certification
C1	Models share dynamical core	$M_{\text{eff}} \ll M$, detection bound degrades
C2	Correlated residual errors	Hoeffding bound invalid; need dependent concentration
C3	Unbounded loss (e.g., raw RMSE)	Concentration inequalities fail
C4	Heterogeneous state errors	Gap Δ_s overestimated; FPR inflated
C5	All models err identically	Noise/bias indistinguishable; Theorem 2 applies
C6	Non-stationary error statistics	Hindcast F1 bound irrelevant for projection
C7	Regime discriminator fails	Novel states misclassified as familiar; overconfidence
C8	Non-isotropic error correlation	SITUS encoding provides no benefit
C9	Non-smooth forcing scenario	Solution uniqueness not guaranteed
C10	Proxy uncertainty too large	Paleoclimate validation inconclusive

11.2 Policy Implications

For climate services that inform adaptation investment (10^{11} – 10^{12} USD/year globally), the difference between “the ensemble mean projects X” and “we certify at confidence $1 - \alpha$ that the true value lies in $[X - \delta, X + \delta]$ ” is operationally enormous. The SCX framework provides the latter form of statement, with the crucial caveat that α and δ may be disappointingly large for many variables of interest—an honest assessment is preferable to a precise but unfounded one.

11.3 Relationship to IPCC Assessment

The IPCC AR6 [6] uses “likelihood” and “confidence” language defined by convention: “likely” means $> 66\%$ probability, “very likely” means $> 90\%$, etc. The SCX framework provides a mathematical foundation for these assignments: under Assumptions C1–C10, the probability bounds in Theorem 1 translate directly to IPCC likelihood categories, with the additional property that the derivation of the bound is fully transparent and reproducible.

11.4 Open Problems

Open Problem 11.1 (Dependent Concentration for Climate Ensembles). Theorem 1 uses Hoeffding’s inequality under Assumption C2 (conditional independence). When C2 is violated, the effective correlation $\bar{\rho}$ degrades M_{eff} . Develop a sharp concentration inequality for dependent Bernoulli trials with known correlation structure that improves the bound beyond the naive variance-inflation factor. ■ [Rigor: SKETCH ★]

Open Problem 11.2 (Optimal Attribution Basis Selection). Given M ESMs with known parameterization differences, what is the information-theoretically optimal decomposition of inter-model spread into an attribution basis $\mathcal{B}$? This is a combinatorial optimization problem over the space of process partitions, coupled to the mutual information between process-resolving observations and model outputs. ■[Rigor: SKETCH ★]

Open Problem 11.3 (Paleoclimate Prior Elicitation). How should paleoclimate constraints be formally incorporated as priors in the Cercis Score? Assumption C10 requires $\sigma_{\text{proxy}} < \tau/2$, but a more sophisticated treatment would use the full proxy likelihood to weight the novelty component N_m , rather than treating paleoclimate as a binary “constrained/unconstrained” gate. ■[Rigor: SKETCH ★]

12 Conclusion

We have presented a mathematical framework for auditing Earth System Model ensembles that provides certified uncertainty bounds rather than informal consensus. The framework rests on 10 explicitly stated assumptions and three theorems with proofs.

Theorem 1 (Systematic Bias Detection) establishes that multi-model consensus provides exponentially convergent detection of systematic biases, with the rate governed by the effective number of independent models M_{eff} . The bound is honest: for the CMIP6 ensemble, certification is modest but real.

Theorem 2 (Attributional Unidentifiability) proves that without declared structural assumptions, it is mathematically impossible to attribute inter-model climate disagreement to specific physical processes. This establishes an epistemic standard: attribution claims must be accompanied by declared and auditable attribution bases.

Theorem 3 (Cercis Consistency) provides a principled ranking of climate projections that balances hindcast skill against the epistemic value of exploring unprecedented regimes. The Cercis Score correctly degrades as models enter novel territory and refuses to rank when no model demonstrates competence.

The SCX framework does not predict the climate. It certifies the epistemic status of predictions made by others. This is a deliberate design choice: certification is separable from modeling, auditable by third parties, and updatable as new models and observations become available. In a domain where trillions of dollars and millions of lives depend on projections that remain fundamentally uncertain, honest certification is the most valuable contribution that mathematical rigor can offer.

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